

WEEK – 4 : Develop Dynamic Webpage Using JavaScript

Aim : To develop a dynamic webpage using JavaScript that accepts student details and subject marks, calculates the total marks, average, grade, and result, and dynamically displays the results on the webpage.

Description : JavaScript can be used to make a webpage dynamic and interactive by responding to user actions and modifying HTML elements without reloading the page.

In this experiment, a Student Marks Calculator is developed using HTML, CSS, and JavaScript. The webpage accepts the student's name and marks in five subjects: English, Maths, Science, Social, and SL.

When the Calculate button is clicked, JavaScript retrieves the values entered by the user and calculates the total and average marks. Based on the average, a grade is assigned from A to D or Fail. The program also checks whether the student has scored at least 35 marks in every subject to determine whether the student has passed or failed.

The webpage uses DOM manipulation and event listeners to dynamically update the result section. The grade color is also changed dynamically according to the grade. The Reset button clears all the entered values and resets the displayed results.

Main Features

- Accepts student's name.
- Accepts marks for five subjects.
- Calculates total marks.
- Calculates average marks.
- Assigns a grade based on average.
- Determines PASS or FAIL.
- Changes the grade color dynamically.
- Displays results without refreshing the webpage.
- Provides a Reset button to clear the form.

Concepts Used

JavaScript Concept	Purpose
getElementById()	Accesses HTML elements
addEventListener()	Handles button click events
Number()	Converts input values into numbers
if-else	Determines grade and result
textContent	Dynamically changes webpage text

.value	Gets or modifies input values
DOM Manipulation	Updates webpage elements dynamically
style.color	Changes the grade color dynamically

Code

HTML Code

```
<!DOCTYPE html>
<html>

<head>
    <title>Student Marks Calculator</title>
    <link rel="stylesheet" href="./style.css">
</head>

<body>

    <h1>Student Marks Calculator</h1>

    <div class="marks">

        <label class="name">Name</label>
        <input
            type="text"
            placeholder="Enter your name"
            class="ans"
            id="studentName"
        >

        <br><br>

        <label class="name">English</label>
        <input
            type="number"
            placeholder="Enter the marks"
            class="ans"
            id="eng"
```

>

<label class="name">Maths</label>

<input

type="number"

placeholder="Enter the marks"

class="ans"

id="math"

>

<label class="name">Science</label>

<input

type="number"

placeholder="Enter the marks"

class="ans"

id="sci"

>

<label class="name">Social</label>

<input

type="number"

placeholder="Enter the marks"

class="ans"

id="soc"

>

<label class="name">SL</label>

<input

```
        type="number"
        placeholder="Enter the marks"
        class="ans"
        id="s1"
    >
```

```
<br><br>
```

```
<button class="btn" id="calculateBtn">
    Calculate
</button>
```

```
<button type="reset" class="btn" id="resetBtn">
    Reset
</button>
```

```
</div>
```

```
<div class="results">
```

```
    <h3>Results:</h3>
```

```
    <p id="student">Name :</p>
```

```
    <p id="total">Total :</p>
```

```
    <p id="average">Average :</p>
```

```
    <p id="grade">Grade :</p>
```

```
    <p id="result">Result :</p>
```

```
</div>
```

```
<script src="./script.js"></script>
```

```
</body>
```

```
</html>
```

CSS Code:

```
body {  
    background-color: blanchedalmond;  
}  
  
/* Heading */  
h1 {  
    display: flex;  
    justify-content: center;  
    margin: 0 auto 20px auto;  
    background-color: white;  
    width: 500px;  
    flex-wrap: wrap;  
    border-radius: 20px;  
    padding: 10px;  
    box-shadow: 0px 10px 20px rgba(0, 0, 0, 0.2);  
}  
  
/* Marks Section */  
.marks {  
    background-color: rgb(199, 167, 230);  
    margin: 80px;  
    padding: 40px;  
    border-radius: 15px;  
    display: flex;  
    flex-direction: column;  
    justify-content: center;  
    gap: 10px;  
}  
  
/* Labels */  
.name {  
    color: rgb(19, 17, 173);  
    font-size: 1rem;  
}
```

```
/* Input Fields */
.ans {
    border-radius: 5px;
    margin: 10px;
    padding: 8px;
    width: 250px;
}

/* Buttons */
.btn {
    width: 100px;
    display: flex;
    border-color: none;
    background-color: rgb(106, 106, 226);
    border-radius: 8px;
    color: white;
    padding: 8px;
    box-shadow: 0px 10px 20px rgba(0, 0, 0, 0.15);
}

.btn:hover {
    transform: scale(1.2);
    transition: 1s;
}

/* Results Section */
.results {
    background-color: rgb(199, 167, 230);
    margin: 80px;
    padding: 40px;
    border-radius: 15px;
    display: flex;
    flex-direction: column;
    justify-content: center;
    gap: 10px;
}
```

```
/* Results Heading */
.results h3 {
  color: white;
  display: flex;
  justify-content: center;
  font-size: 1.5rem;
}
```

```
/* Results Paragraphs */
.results p {
  color: rgb(72, 64, 190);
  font-weight: bold;
}
```

Javascript code:

```
const calculateBtn = document.getElementById("calculateBtn");
```

```
calculateBtn.addEventListener("click", function () {
```

```
  // Get student name
```

```
  const studentName =
```

```
    document.getElementById("studentName").value;
```

```
  document.getElementById("student").textContent =
```

```
    `Name : ${studentName}`;
```

```
  // Get marks
```

```
  const english =
```

```
    Number(document.getElementById("eng").value);
```

```
  const maths =
```

```
    Number(document.getElementById("math").value);
```

```
  const science =
```

```
    Number(document.getElementById("sci").value);
```

```
const social =
    Number(document.getElementById("soc").value);

const sl =
    Number(document.getElementById("sl").value);

// Calculate total
const total =
    english + maths + science + social + sl;

document.getElementById("total").textContent =
    `Total : ${total}`;

// Calculate average
const average = total / 5;

document.getElementById("average").textContent =
    `Average : ${average}`;

// Calculate grade
let grade;

if (average >= 90) {
    grade = "A";
}
else if (average >= 80) {
    grade = "B";
}
else if (average >= 70) {
    grade = "C";
}
else if (average >= 60) {
    grade = "D";
}
else {
    grade = "Fail";
}
```

```
}

const gradeElement =
    document.getElementById("grade");

gradeElement.textContent =
    `Grade : ${grade}`;

// Change grade color
if (grade === "A") {
    gradeElement.style.color = "green";
}
else if (grade === "B") {
    gradeElement.style.color = "blue";
}
else if (grade === "C") {
    gradeElement.style.color = "yellow";
}
else if (grade === "D") {
    gradeElement.style.color = "orange";
}
else {
    gradeElement.style.color = "red";
}

// Calculate result
let result;

if (
    english >= 35 &&
    maths >= 35 &&
    science >= 35 &&
    social >= 35 &&
    sl >= 35
) {
    result = "PASS";
}
```

```
}
else {
    result = "FAIL";
}

document.getElementById("result").textContent =
    `Result : ${result}`;
});

// Reset Button
const resetBtn = document.getElementById("resetBtn");

resetBtn.addEventListener("click", function () {

    document.getElementById("studentName").value = "";
    document.getElementById("eng").value = "";
    document.getElementById("math").value = "";
    document.getElementById("sci").value = "";
    document.getElementById("soc").value = "";
    document.getElementById("sl").value = "";

    document.getElementById("student").textContent =
        "Name :";

    document.getElementById("total").textContent =
        "Total :";

    document.getElementById("average").textContent =
        "Average :";

    document.getElementById("grade").textContent =
        "Grade :";

    document.getElementById("result").textContent =
        "Result :";
```

```
document.getElementById("grade").style.color = "";  
});
```

Output :

Student Grade calculator

Name

English

Maths

Science

Social

PE

Calculate

Reset

Results:

Points : 528

Total : 420

Average : 84.8

grade : B

Result : 84.8